Research & Other Activities Report (Mar - Q3 2025)

During Winter,March // Other Details

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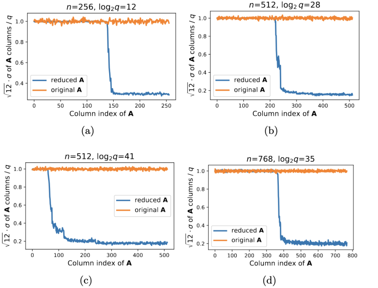
Outline

1. Research Progress
   - Current Focus
   - Formal Model
   - Outline
2. Job-Hunting Update
   - NTT R&D Track
   - Other Career Paths
3. Miscellaneous
   - Scholarships & Courses

Binary-LWE Key-Recovery Project

- **Goal:** Efficient recovery of sparse binary secrets in LWE via an extended *Cool-Chill-Cruel* hybrid attack.
- **Recap:** Exploit *intermediate-variance ("Chill")* columns instead of ignoring them. $\Rightarrow$ smaller brute-force core, maintain success probability.
- **Status (Start – Q2 FY 2025):**
  1. Formal model drafted; complexity bound derived.
  2. Lattice-reduction parameter sweep scripted (BKZ / flatter).
  3. ML/light-statistic classifiers on Chill windows pending benchmarking; expecting $k = 4$-64.
- **Next 8 weeks:**
  - Finish variance-profiling pipeline; publish dataset.
  - Integrate logistic baseline vs. tiny-Transformer/sc head.
  - Draft workshop abstract if time allows.

Cool-Chill-Cruel   Formal Snapshot



Empirical st. dev. profile (ex. $n\!=\!512$, $\log_2 q\!=\!41$): –75 $=$ *Cruel*, 75–115 $=$ *Chill*, 116– $=$ *Cool*. A visible "slope" motivates the extra middle zone. Chill reg. is up to 40b.

Column zoning after partial lattice reduction

$$T_c > T_h, \quad \sigma_j = \mathrm{stdev}(A'_{*,j})$$

$$\mathrm{Region}(j) = \begin{cases} \textit{Cruel} & \sigma_j \geq T_c \\ \textit{Chill} & T_h \leq \sigma_j < T_c \\ \textit{Cool} & \sigma_j < T_h \end{cases}$$

Key numbers (typ. 512-dim):

$$\rho \approx 0.41$$

$$n_{\text{cruel}} \approx 75 - 190$$

$$n_{\text{chill}} \approx 4 - 40$$

Broader Plan & PhD Option

- Continue master's thesis path, **but** preparing PhD application (domestic, Japan) as primary (Consider DC1, outsource employment, etc).
- Potential PhD topic extension:
  1. Ring-LWE rotation / leakage.
  2. Adaptive defense: secret-distribution hardening.

NTT R&D Application

- Online entry sheet drafted; internal review this week.
- Next steps:
  1. Submit by **end of Apr**.
  2. Expect aptitude + technical interview.

Additional Employment Activities

- **Current employer (remote dev):** negotiating long-term outsourcing contract through graduation; discussing possible business trips to US.
- Preparing 2-3 more applications (possibly consultancies / startups).
- Networking via conferences + LinkedIn JP market groups.

Other Commitments

- **Scholarships 2025:** reviewing private foundations (noted on several misleading conditions).
- **Course load (Semester):**
  - Cryptography, Soft Computing
  - Japanese: upperintermediate/advanced, communication for business purposes
- **JLPT N1** scheduled July; intensive prep.

Take-aways

- Research: hybrid *Cool-Chill-Cruel* attack progressing; variance study + ML classifier next.
- Career: NTT R&D application in final stage; parallel options open.
- Personal development: scholarship search, JLPT N1, 4 graduate courses.

Q?